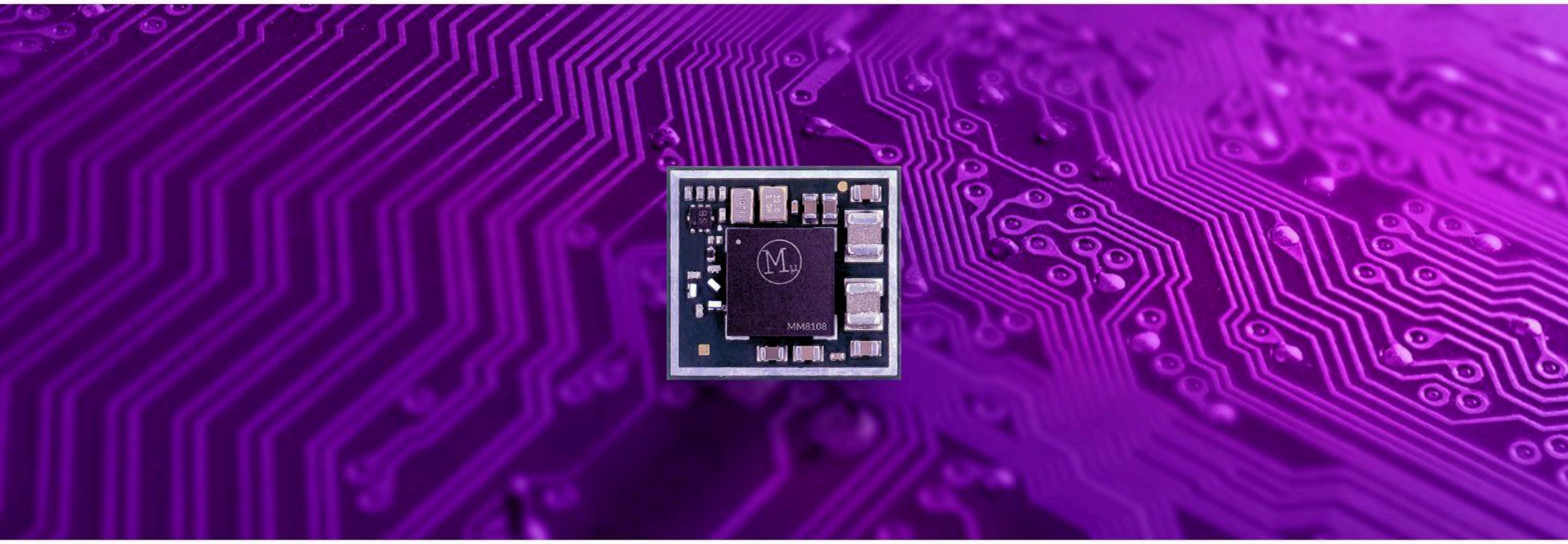
Morse Micro



MM8108-MF15457
Data Sheet

Multi-region Wi-Fi HaLow/IEEE 802.11ah Module
with USB Host Interface and Integrated Amplifiers

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Refer to the official digital version for the latest updates

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1 Product Overview

1.1 Introduction

Wi-Fi HaLow (pronounced "HEY-low") is the first global Wi-Fi standard (IEEE 802.11ah) specifically designed to meet the needs of the Internet of Things (IoT). It's an open standard wireless network technology operating in the sub-1 GHz license-exempt RF bands (850-950 MHz range), so it doesn't incur ongoing monthly costs like cellular/mobile network connections. By operating in the sub-1 GHz range, this low-power wireless protocol can connect more IoT devices over much longer distances and with significantly lower power than traditional Wi-Fi.

Morse Micro, the world's leading Wi-Fi HaLow solutions provider, offers several Wi-Fi HaLow connectivity solutions. The MM8108-MF15457 is a fully integrated Wi-Fi HaLow module with long range, low power consumption, and excellent RF performance, featuring our second-generation MM8108 Wi-Fi HaLow SoC. It is compliant with IEEE 802.11ah, supporting PHY rates up to 43.3 Mbps, and provides for operation in the sub-1 GHz license-exempt RF bands. The radio in the MM8108 supports programmable operation in these bands worldwide, spanning from 850 MHz to 950 MHz.

The MM8108's integrated power amplifier (PA) offers world-leading transmit performance and energy efficiency with on-chip amplification of up to 26 dBm, which is ideal for low-cost devices. The ultra-high linearity integrated low-noise amplifier (LNA) offers exceptional receive sensitivity. These innovations, combined with digital transmit filtering and pre-distortion circuits, result in an advanced transceiver design that ensures out-of-the-box compatibility with all global Wi-Fi HaLow regulatory environments.

MM8108-MF15457 supports a USB 2.0 High-Speed host interface and an SDIO/SPI host interface, offering solution architects flexibility when integrating Wi-Fi HaLow connectivity into their existing solution.

With a footprint of only 11 mm x 10 mm, the MM8108-MF15457 module has been designed to showcase the cost and size benefits of the higher level of integration of our second-generation SoC.

MM8108-MF15457 supports WPA3 and all standard security features required for Wi-Fi HaLow product certifications.

1.2 Features

- Single-stream maximum PHY rate of 43.3 Mbps at 8 MHz bandwidth
- Radio supporting worldwide Sub-1 GHz frequency bands
  - Frequency range: 850-950 MHz
  - Channel bandwidth options of 1/2/4/8 MHz
  - High power and extremely high efficiency integrated PA with maximum 26 dBm (400 mW) output power at 45% PA efficiency and 37% SoC efficiency
  - Novel PA architecture maintains efficiency at power backoff
  - Low-power receiver with integrated LNA, NF < 4 dB
- 802.11ah OFDM PHY with Wi-Fi Alliance Wi-Fi HaLow certification
  - BPSK & QPSK, 16-QAM, 64-QAM and 256-QAM Modulation
  - Automatic frequency and gain control
  - Packet detection and channel equalization
  - Forward Error Correction (FEC) coding and decoding
  - Modulation and Coding Scheme (MCS) levels 0-10
  - 1 MHz duplicate mode
  - Optional traveling pilots
- 802.11ah MAC with Wi-Fi Alliance Wi-Fi HaLow certification
  - Support for Station (STA) and Access Point (AP) roles
  - Listen-Before-Talk (LBT) access with energy detection
  - 802.11 power save
  - 802.11 fragmentation and defragmentation
  - Power-Saving Target Wake Time (TWT) support for extended battery life
  - Automatic and manual MCS rate selection
- On-chip software stack assisting with host offload of Wi-Fi connection management
- USB 2.0 High-Speed compliant device interface
  - Integrated USB 2.0 PHY and device interface supporting High-Speed mode at 480 Mbps
- SDIO 2.0 compliant device interface
  - SDIO 2.0 High-Speed at 50 MHz max for 200 Mbps
  - Support for 1-bit and 4-bit data mode
  - Support for SPI mode operation at up to 80 MHz for 80 Mbps
- Power Management Unit (PMU) supporting various modes of operation
  - Single supply 3.0-3.6 V for integrated DC-DCs and LDOs
  - Multiple low-power modes (hibernate, deep sleep, snooze) support extreme power environments
- Broad spectrum of security features
  - Wi-Fi layer security, including Wi-Fi Protected Access version 3 (WPA3), Protected Management Frames (PMF), and Opportunistic Wireless Encryption (OWE)
  - Hardware support for Advanced Encryption Standard (AES) and Secure Hash Algorithm version 2 (SHA-2) functions (SHA-256, SHA-384, SHA-512)

1.3 Applications

Ideally suited for Internet of Things (IoT) and Machine-to-Machine (M2M) applications such as:

- Surveillance cameras and sensors
- Cloud connectivity
- Low-power sensor networks
- Building automation systems
- Asset tracking and management
- Machine performance monitors and sensors
- Building access control and security
- Drone video and navigation communications
- Connected toys and games
- Rural internet access
- Agricultural networks
- Utility smart meter and intelligent grid
- Proximity sensors
- Industrial automation controls
- Smart home automation
- EV car chargers
- Appliances
- Construction site connectivity
- Smart signs and kiosks
- Retail point-of-sale terminals
- Vehicle-to-vehicle communications
- IP sensor networks
- Biometric IDs and keypads
- Warehouse connectivity
- Intelligent lighting controls
- BT/ZigBee™/Z-Wave™ to Wi-Fi HaLow gateways
- Wi-Fi to Wi-Fi HaLow bridges
- Wi-Fi HaLow client adapters/dongles
- Smart city networks

2 Pin Descriptions

The MM8108-MF15457 module features 38 pins, which are described in this section. The following illustration shows the top view of the module pins.

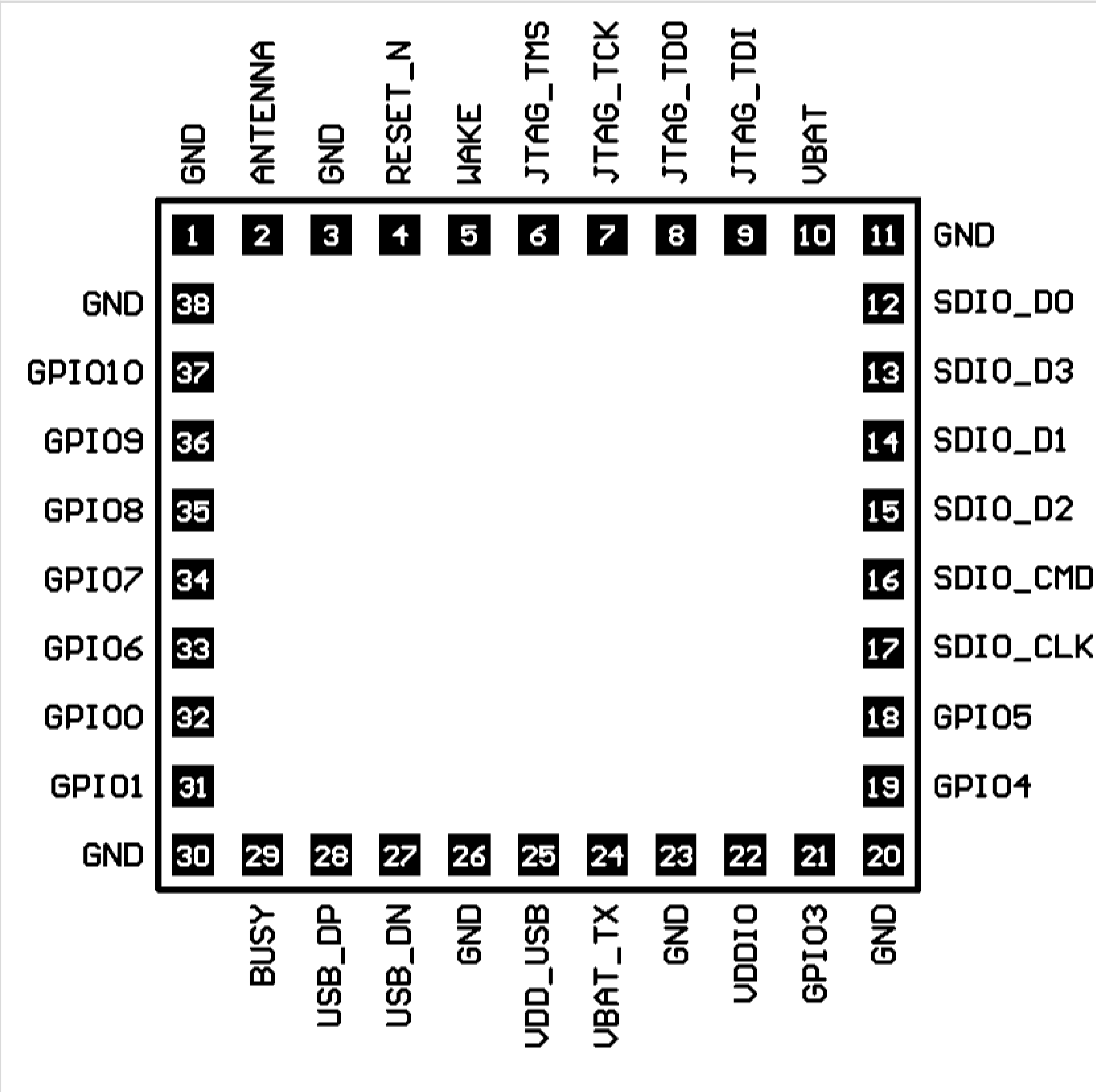

Figure 1: Pin diagram

Pin	Pin Name	Type	Description	Alternative
1	GND	Ground	Ground	
2	ANT	Analog	Antenna	
3	GND	Ground	Ground	
4	RESET_N[2]	Digital I/O	Asynchronous chip reset (active low)	
5	WAKE[2]	Digital I/O	External wake from Deep Sleep and Snooze	
6	JTAG_TMS	Digital I/O	JTAG Mode Select	GPIO15[3]
7	JTAG_TCK	Digital I/O	JTAG Clock	GPIO13[3]
8	JTAG_TDO	Digital I/O	JTAG Data Out	GPIO16[3]
9	JTAG_TDI	Digital I/O	JTAG Data In	GPIO14[3]
10	VBAT	Supply	3.3 V VBAT Supply	
11	GND	Ground	Ground	
12	SDIO_D0[1]	Digital I/O	SDIO Data line	SPI_MISO
13	SDIO_D3[1]	Digital I/O	SDIO Data line	SPI_CS
14	SDIO_D1[1]	Digital I/O	SDIO Data line	SPI_INT
15	SDIO_D2[1]	Digital I/O	SDIO Data line	
16	SDIO_CMD[1]	Digital I/O	SDIO Command line	SPI_MOSI
17	SDIO_CLK	Digital I/O	SDIO Clock input	SPI_SCK
18	GPIO5[3]	Digital I/O	Programmable digital I/O	
19	GPIO4[3]	Digital I/O	Programmable digital I/O	
20	GND	Ground	Ground	
21	GPIO3[3]	Digital I/O	Programmable digital I/O	
22	VDDIO	Supply	Host supply for digital I/O	
23	GND	Ground	Ground	
24	VBAT_TX	Supply	3.3 V VBAT-TX Supply	
25	VDD_USB	Supply	USB Supply	
26	GND	Ground	Ground	
27	USB_D_N	Digital I/O	USB DM line	
28	USB_D_P	Digital I/O	USB DP line	
29	BUSY	Digital I/O	BUSY signal output	
30	GND	Ground	Ground	
31	GPIO1[3]	Digital I/O	Programmable digital I/O	
32	GPIO0[3]	Digital I/O	Programmable digital I/O	
33	GPIO6[3]	Digital I/O	Programmable digital I/O	
34	GPIO7[3]	Digital I/O	Programmable digital I/O	
35	GPIO8[3]	Digital I/O	Programmable digital I/O	
36	GPIO9[3]	Digital I/O	Programmable digital I/O	
37	GPIO10[3]	Digital I/O	Programmable digital I/O	
38	GND	Ground	Ground	

[1] All SDIO bus pins except SDIO_CLK should be pulled up with a 10 kΩ to 100 kΩ resistor as per the SDIO standard.

[2] Supplied from VBAT domain. The VDDIO domain drives other digital pins.

[3] GPIOs 0 to 10 may provide alternative functions, including SPI master, I2C master, PWM, and UART. This functionality necessitates custom software, which Morse Micro can develop upon request. For further details, please contact Morse Micro.

3 Functional Description

The following sections describe the functions of the MM8108-MF15457 module.

3.1 Block Diagram

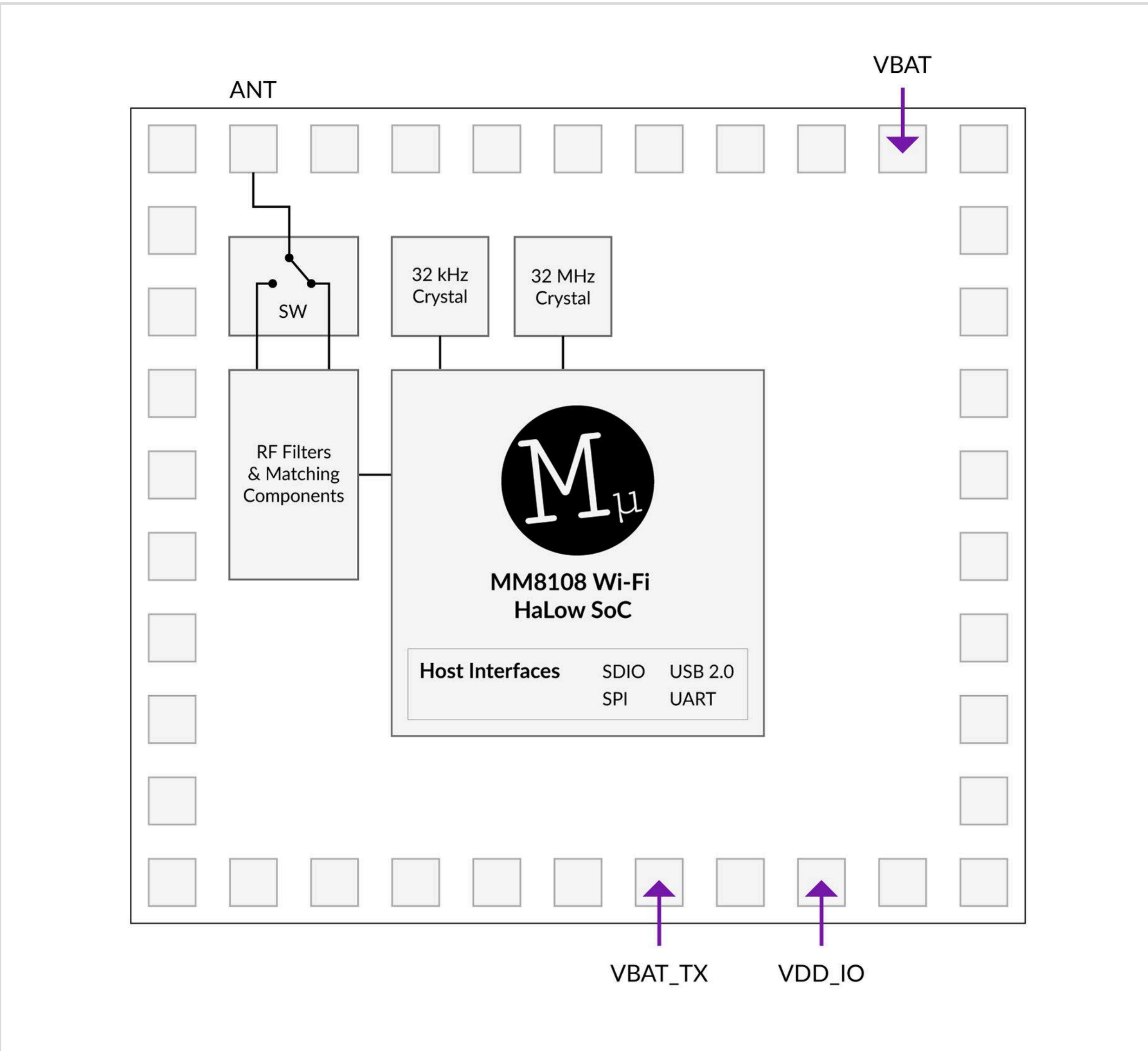

Figure 2: Functional block diagram

3.2 Power Supply Requirements

MM8108-MF15457 module power is derived from a 3.0 to 3.6 V supply on pins VBAT and VBAT_TX.

VDDIO sets the IO voltage of the MM8108-MF15457. It has an input voltage range of 2.25 V to 3.6 V and should be connected to the same power supply as the host MCU.

There are no strict power-up sequencing requirements.

3.3 USB Host Interface

A schematic diagram detailing the USB host interface circuit is shown below:

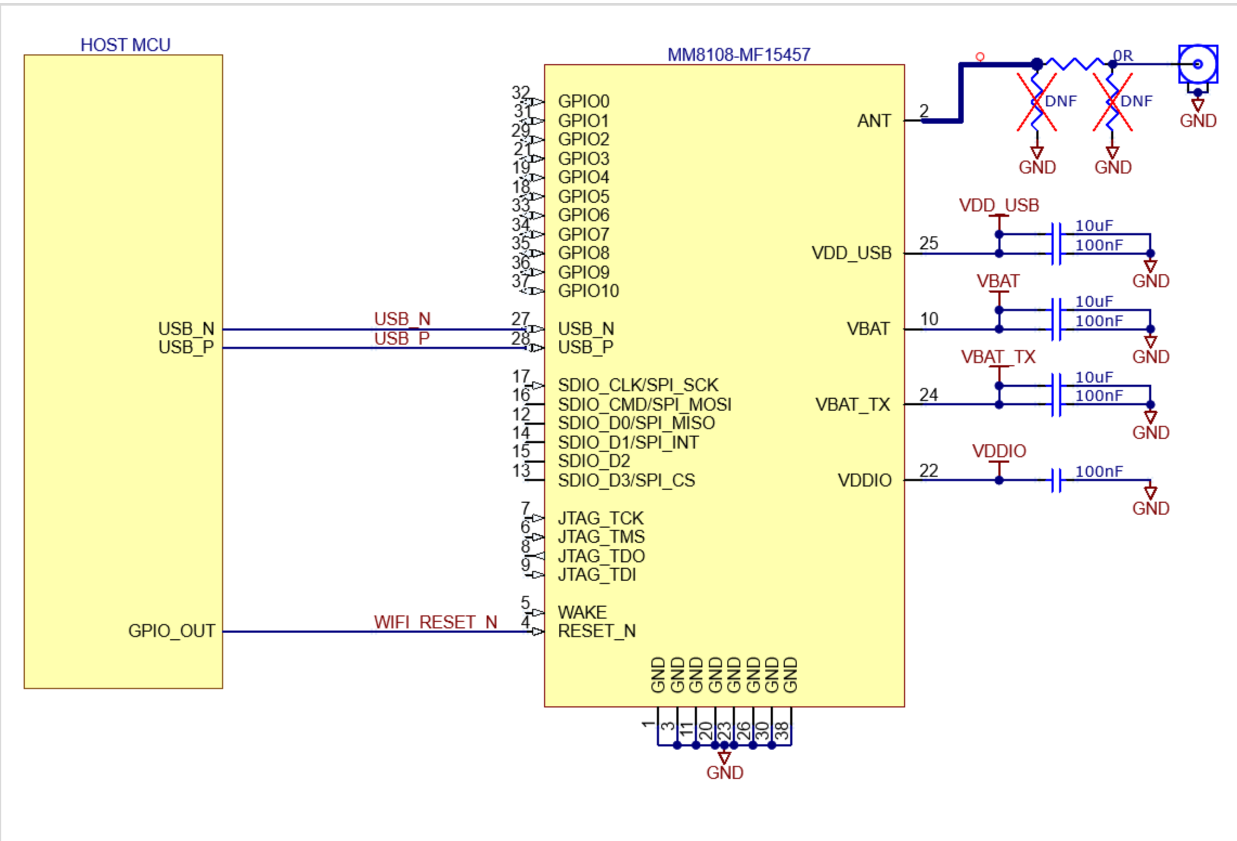

Figure 3: USB host interface circuit

3.4 USB Network Adapter Interface

A schematic diagram detailing the USB network adapter circuit is shown below:

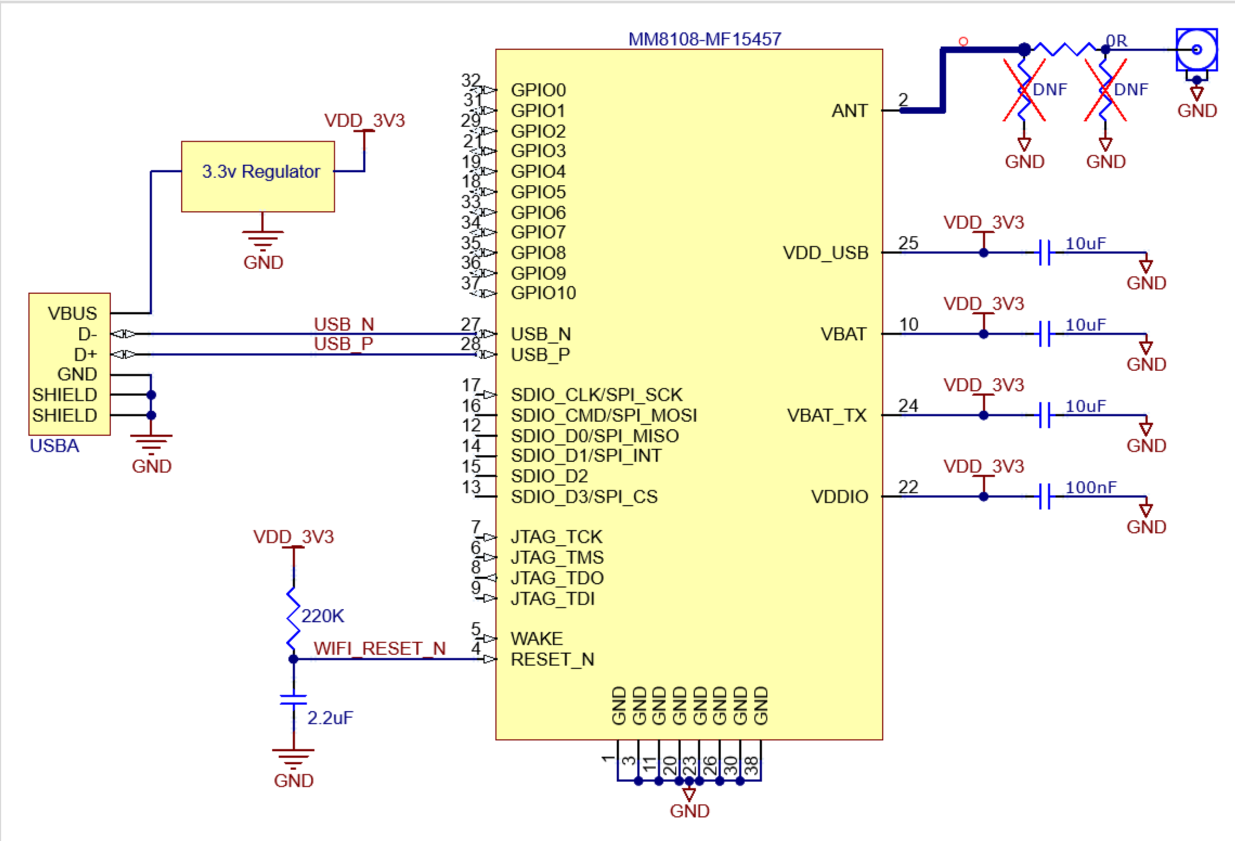

Figure 4: USB network adapter circuit

3.5 SDIO Host Interface

When selecting a host to interface with the MM8108-MF15457 module via the SDIO interface, ensure the host supports SDIO 2.0 with SDIO clock speeds of up to 50 MHz. Slower clock speeds will impact the maximum achievable throughput.

The SDIO data and command lines should be pulled up with 10 kΩ to 100 kΩ resistors per the SDIO 2.0 specification.

For proper operation and to take advantage of the module's power-saving features, connect RESET_N and WAKE to standard digital outputs (CMOS logic levels). The BUSY signal should be connected to a digital input (also CMOS logic levels). Do not use open-collector or open-drain circuits, as they will cause incorrect behavior.

In applications where the module must always be on, and the power-saving features can not be used, such as access points, the WAKE pin can be left not connected, reducing the need for GPIOs on the host processor to only one.

A schematic diagram detailing the SDIO host interface circuit using the module's power-saving features is shown below:

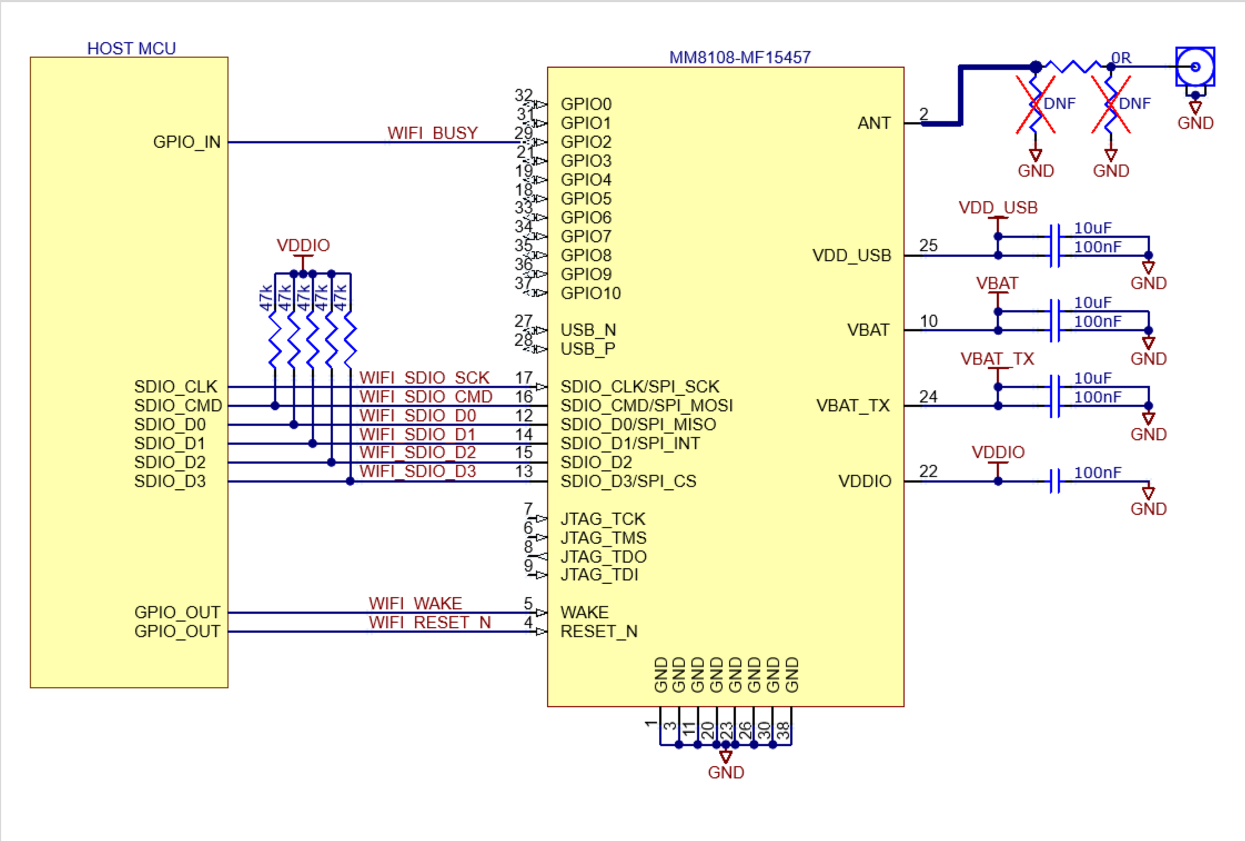

Figure 5: SDIO host interface circuit using power-saving features

A schematic diagram detailing the SDIO host interface circuit for always-on applications is shown below:

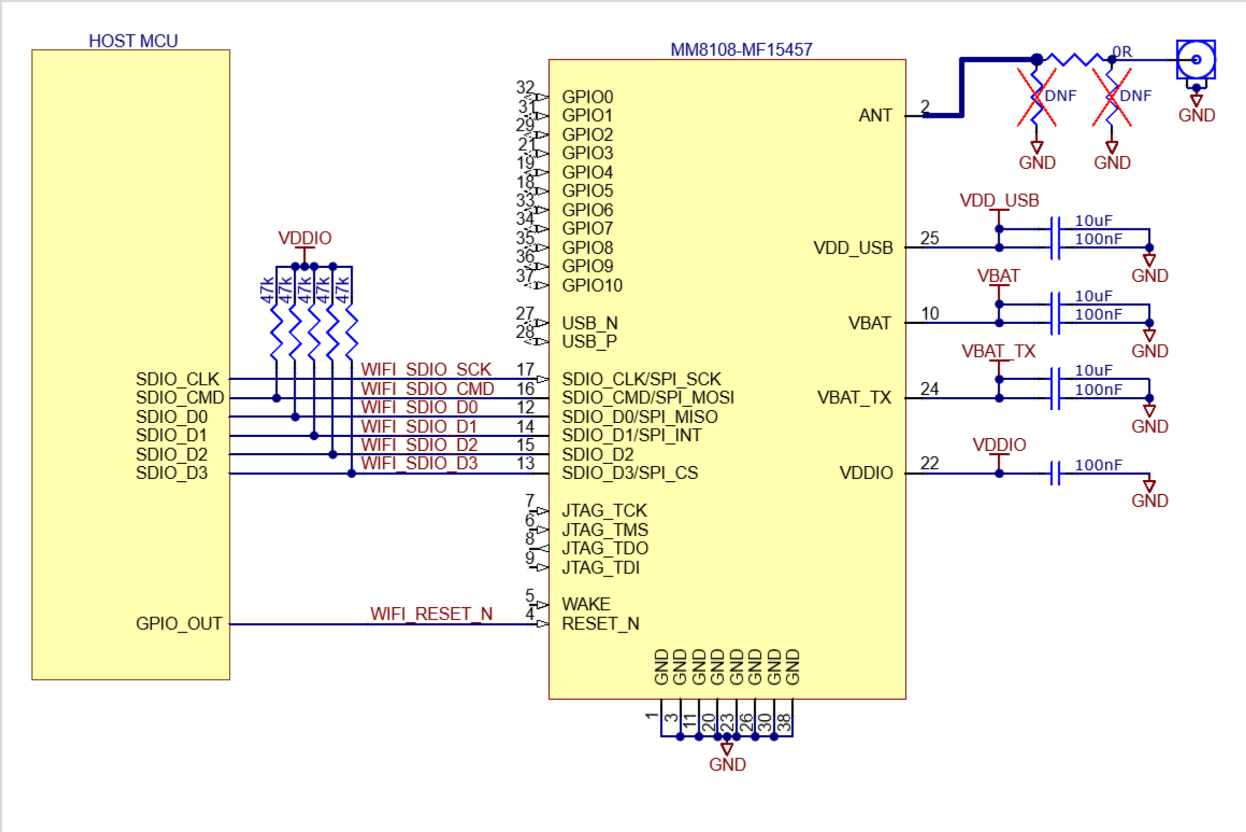

Figure 6: SDIO host interface circuit for always-on applications

3.6 SPI Host Interface

When selecting a host to interface with the MM8108-MF15457 module via the SPI interface, consider the following recommendations to achieve the best throughput:

- The host must support level-triggered interrupts.
- The host must support full-duplex SPI mode.
- The host must support DMA-backed transactions on the SPI bus.

Standard SPI can achieve up to 25 Mbps at 50 MHz, which will be significantly reduced without DMA support. For example, an SPI interface with an 8-byte buffer per transaction might only achieve 2 Mbps throughput on the SPI bus.

For proper operation and to take advantage of the module's power-saving features, connect RESET and WAKE to standard digital outputs (CMOS logic levels). The BUSY signal should be connected to a digital input (also CMOS logic levels). Do not use open-collector or open-drain circuits, as they will cause incorrect behavior.

In applications where the module must always be on, and the power-saving features can not be used, such as access points, the WAKE pin can be left not connected, reducing the need for GPIOs on the host processor to only one.

A schematic diagram detailing the SPI host interface circuit using the module's power-saving features is shown below:

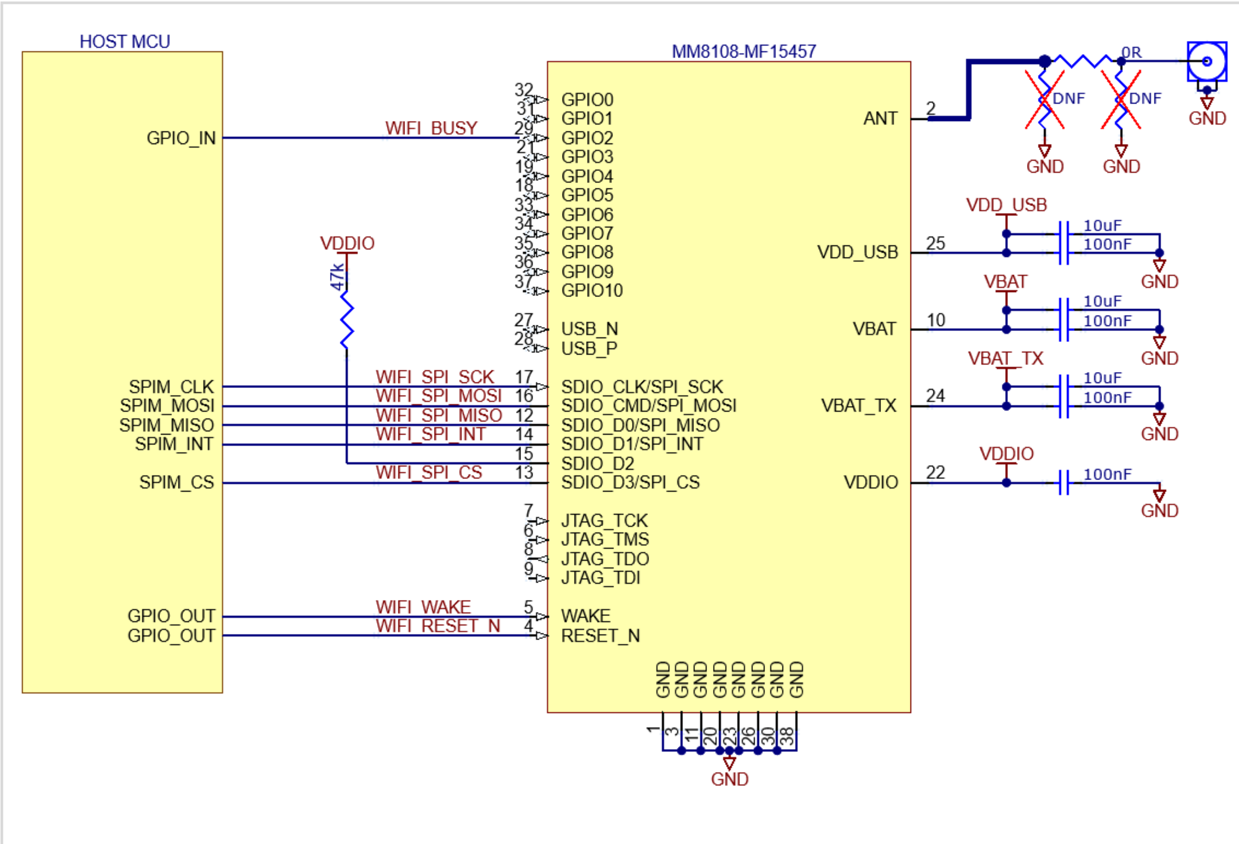

Figure 7: SPI host interface circuit using power-saving features

A schematic diagram detailing the SPI host interface circuit for always-on applications is shown below:

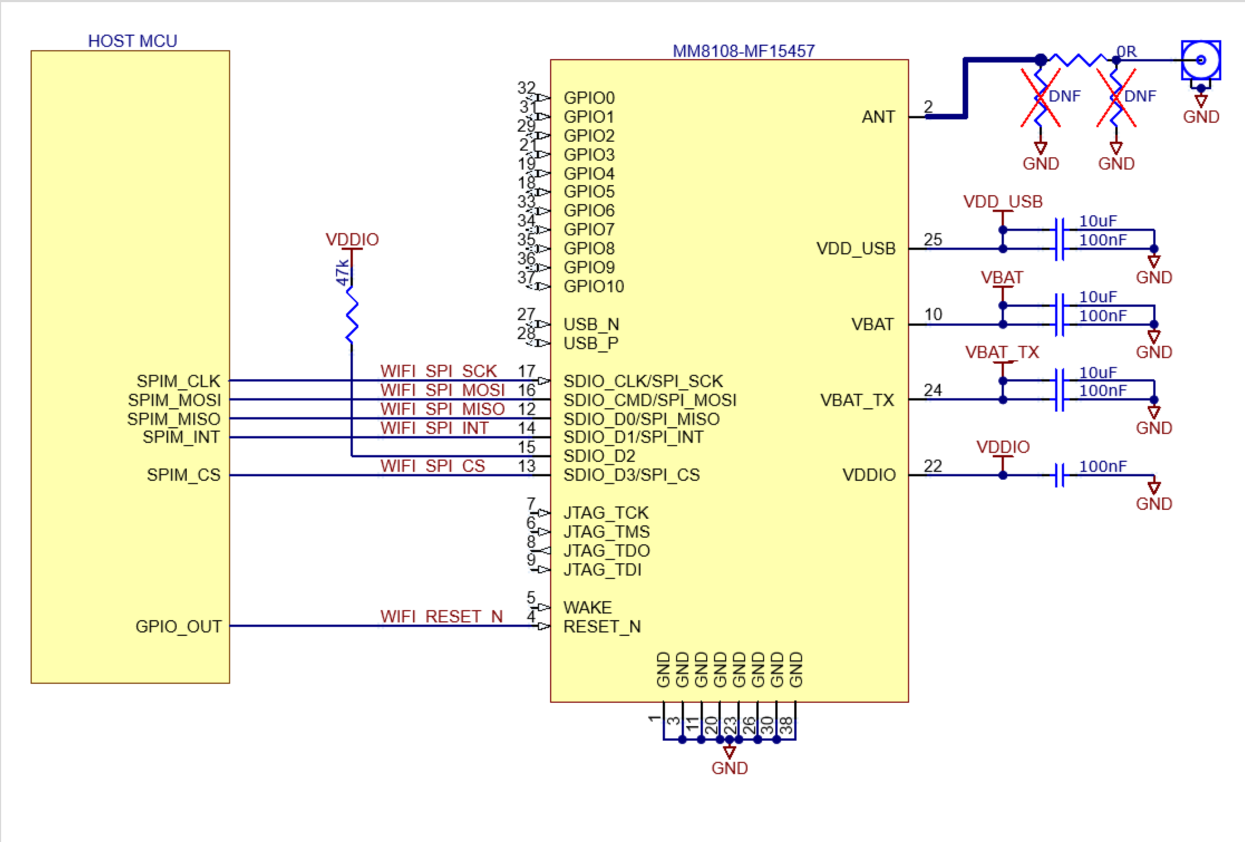

Figure 8: SPI host interface circuit for always-on applications

3.7 Boot/Reset Sequencing

3.7.1 Boot Timing

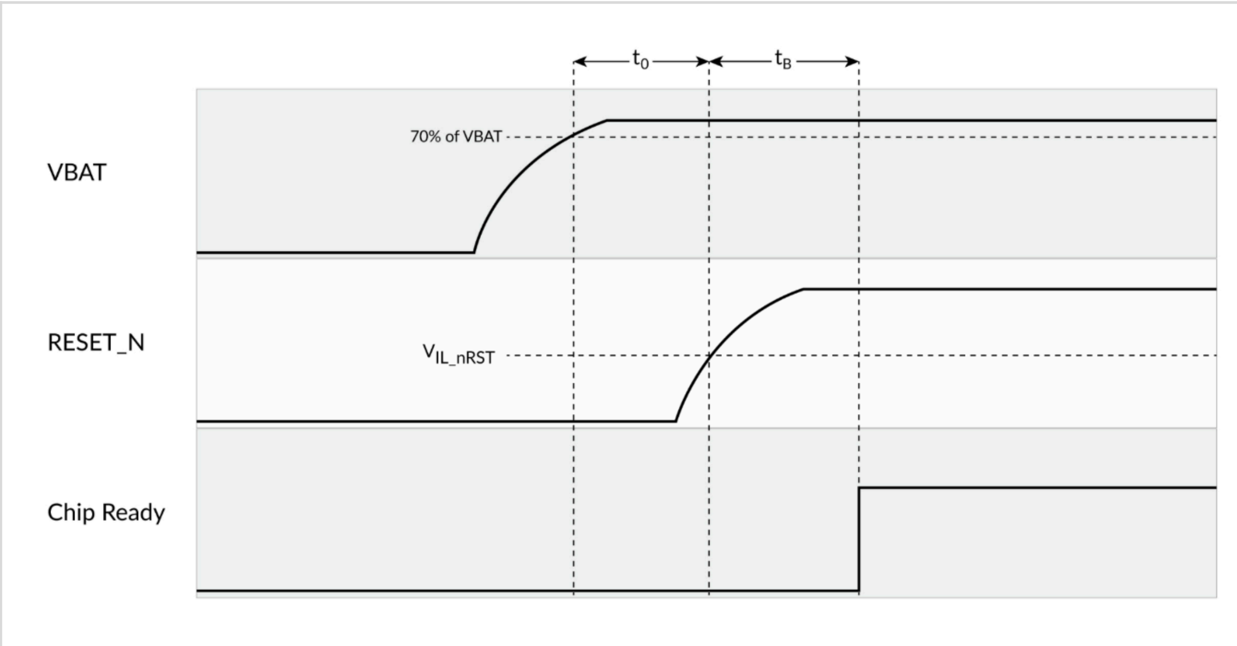

Figure 9: Powering on and reset timing diagram

Parameters	Description	Min	Max	Unit
V_{IL_nRST}	Reset threshold	450		mV
t_0	Time between VBAT brought up (3.3V) and RESET_N being activated	50		µs
t_B	Boot Time		10	ms

Table 1: Boot timing overview

3.7.2 Reset Timing

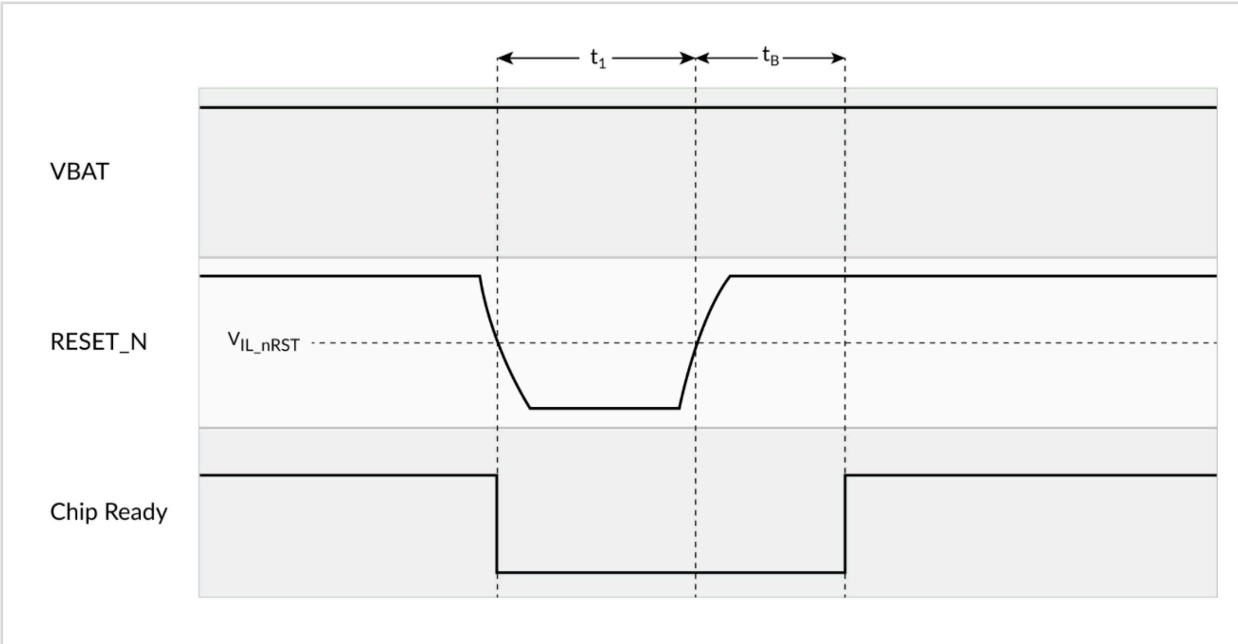

Figure 10: Powering on and reset timing diagram

Parameters	Description	Min	Max	Unit
V_{IL_nRST}	Reset threshold	450		mV
t_1	Duration of RESET_N signal level < VIL_nRST to reset the chip	1000		µs
t_B	Boot Time		10	ms

Table 2: Reset timing overview

4 Electrical Characteristics

4.1 Absolute Maximum Ratings

Stress beyond absolute maximum ratings may cause permanent damage to the MM8108-MF15457 module. Operation is only guaranteed within the recommended operation conditions. Operation of the device outside of the recommended conditions may result in a reduced lifetime and reliability problems, even if the absolute maximum ratings are not exceeded.

Parameter	Min	Max	Unit
VBAT voltage	-0.3	3.6	V
VBAT_TX voltage	-0.3	3.6	V
RESET_N/WAKE	-0.3	3.6	V
VDDIO	-0.3	3.6	V
Analog/RF pin	-0.3	1.0	V
Storage temperature	-40	125	°C
RF input power (CW)	-	6	dBm

Table 3: Absolute maximum and minimum voltage ratings

4.2 ESD Immunity

<table>
<tr><th colspan="3">Parameter</th><th>Min</th><th>Max</th><th>Unit</th></tr>
<tr><td rowspan="3">Electrostatic discharge (ESD) performance</td><td rowspan="2">Human body model (HBM), per ANSI / ESDA / JEDEC JS001</td><td>RF Input</td><td>-1,000</td><td>1,000</td><td>V</td></tr>
<tr><td>All pins except RF Input</td><td>-2000</td><td>2000</td><td>V</td></tr>
<tr><td>Charged device model (CDM), per JESD22-C101</td><td>All pins</td><td>-500</td><td>500</td><td>V</td></tr>
</table>

Table 4: ESD immunity specifications

4.3 Recommended Operating Conditions

Parameter	Min	Typ	Max	Unit
Ambient temperature	-40		85	°C
Storage temperature	-40		125	°C
VBAT	3.0	3.3	3.6	V
VBAT_TX	3.0	3.3	3.6	V
VDDIO	2.25	3.3	3.6	V

Table 5: Recommended operating conditions

4.4 Power Consumption

4.4.1 Active Transmit Current Consumption

MCS Index	Current consumption (mA) per BW at V_{BAT}				Current consumption (mA) per BW at V_{BAT_TX}			
	100% duty cycle, T_A=25ºC, V_{BAT}=V_{DDIO}=3.3 V							
	1 MHz	2 MHz	4 MHz	8 MHz	1 MHz	2 MHz	4 MHz	8 MHz
10	51	N/A	N/A	N/A	281	N/A	N/A	N/A
0	52	51	50	59	279	243	160	132
1	52	51	50	59	278	243	159	131
2	51	51	50	59	277	243	158	130
3	51	51	50	59	266	242	158	130
4	49	50	50	59	221	221	152	130
5	44	47	48	58	144	174	128	125
6	42	44	46	57	118	119	109	106
7	36	39	44	54	88	96	98	95
8	34	38	43	52	69	78	83	81
9	32	N/A	40	49	54	N/A	63	58

Table 6: Transmit current consumption for VBAT, VBAT_TX separated

4.4.2 Active Receive Current Consumption

MCS Index	V_{BAT} + V_{BAT_TX} + V_{DDIO} Current consumption (mA) per BW			
	100% duty cycle, T_A=25ºC, V_{BAT}=V_{DDIO}=3.3 V, Rx signal at -70dBm			
	1 MHz	2 MHz	4 MHz	8 MHz
0	20	21	24	32
Max	20 (MCS9)	21 (MCS8)	26 (MCS9)	34 (MCS9)

Table 7: Active receive current consumption

4.4.3 Listen Receive Current Consumption

V_{BAT} + V_{BAT_TX} + V_{DDIO} Current consumption (mA) per BW			
T_A=25°C, V_{BAT}=V_{DDIO}=3.3 V, Rx signal at -70dBm			
1 MHz	**2 MHz**	**4 MHz**	**8 MHz**
19	20	23	29

Table 8: Listen receive current consumption

4.4.4 Low Power Current Consumption

Power Mode	Current Consumption		
	T_A=25°C, V_{BAT}=V_{DDIO}=3.3 V		
	V_{BAT} + V_{BAT_TX}	V_{DDIO}	Unit
Hibernate	<1	<1	µA
Deep Sleep	<2	<1	µA
Snooze	<20	<1	µA
USB snooze	<500	<1	µA

Table 9: Low power current consumption

4.4.5 Standby Current Consumption (using SDIO/SPI Host Interface)

Mode	Condition: T_A=25°C, V_{BAT}=V_{DDIO}=3.3 V	V_{BAT} + V_{BAT_TX}	V_{DDIO}	Unit
DTIM3	2MHz channel, RTC=RTC_XTAL, beacon=long	176	<1	µA
DTIM10	2MHz channel, RTC=RTC_XTAL, beacon=long	68	<1	µA
DTIM40	2MHz channel, RTC=RTC_XTAL, beacon=long	33	<1	µA
DTIM3	2MHz channel, RTC=RTC_XTAL, beacon=short	126	<1	µA
DTIM10	2MHz channel, RTC=RTC_XTAL, beacon=short	52	<1	µA
DTIM40	2MHz channel, RTC=RTC_XTAL, beacon=short	29	<1	µA

Table 10: Standby current consumption

The DTIM (Delivery Traffic Indication Message) number (DTIM**x**) indicates how often a Wi-Fi HaLow station in power-saving Snooze mode should wake to receive traffic from the access point. It is expressed as a multiple of beacon intervals (multiples of 102.4 ms).

For example, DTIM1 indicates that the station should wake up every 102.4 ms, while DTIM40 configures the station to wake up every 4.1 seconds.

Configuring DTIM is a complex compromise between power consumption and latency. Increasing the DTIM will reduce the average power consumption, as the station spends a greater proportion of time in Snooze mode, which consumes significantly less power than in Listen mode. However, since the station is unable to receive traffic in Snooze mode, increasing DTIM also increases latency.

DTIM settings should be optimized appropriately according to the use case.

4.5 RF Specifications

4.5.1 Frequency Range

The MM8108-MF15457 radio operates in the frequency range from 850 MHz to 950 MHz.

4.5.2 Receiver

Sensitivities for 10% packet error rate, 256-byte packets. Conditions are VBAT=3.3 V, VBAT_TX=3.3 V, 25°C

MCS index	Modulation scheme	Coding rate	PHYrate (Mbps) per BW				Minimum receive sensitivity (dBm) per BW			
			1 MHz	2 MHz	4 MHz	8 MHz	1 MHz	2 MHz	4 MHz	8 MHz
10	BPSK	1/2 x 2	0.1	N/A	N/A	N/A	-107	N/A	N/A	N/A
0	BPSK	1/2	0.3	0.7	1.5	3.3	-106	-103	-102	-98
1	QPSK	1/2	0.7	1.4	3.0	6.5	-104	-101	-99	-95
2	QPSK	3/4	1.0	2.2	4.5	9.8	-102	-99	-97	-93
3	16-QAM	1/2	1.3	2.9	6.0	13	-99	-96	-94	-90
4	16-QAM	3/4	2.0	4.3	9.0	20	-96	-93	-90	-87
5	64-QAM	2/3	2.7	5.8	12	26	-92	-89	-86	-83
6	64-QAM	3/4	3.0	6.5	14	29	-90	-87	-85	-81
7	64-QAM	5/6	3.3	7.2	15	33	-89	-86	-83	-80
8	256-QAM	3/4	4.0	8.9	18	39	-85	-82	-79	-76
9	256-QAM	5/6	4.4	N/A	20	43	-83	N/A	-78	-74

Table 11: Rate vs receiver sensitivity

4.5.3 Transmitter

Note: The following transmit power levels are for IEEE compliance for 802.11ah. They do not consider any backoffs needed for regional spectrum compliance (e.g., FCC, IC, TELEC). TX power in edge channels will be limited by the restricted band edge in the regulatory regions.

MCS Index	Power output (dBm) at the module ANT pin per BW T_A=25°C, V_{BAT}=V_{DDIO}=3.3 V, IEEE-compliant with 1.5dB SEM margin			
	1MHz	2MHz	4MHz	8MHz
10	25.5	N/A	N/A	N/A
0	25.5	25.0	22.5	22.5
1	25.5	25.0	22.5	22.0
2	25.5	25.0	22.5	22.0
3	25.0	25.0	22.5	22.0
4	24.0	24.5	22.5	22.0
5	21.5	23.0	21.5	21.5
6	22.5	21.5	20.5	20.5
7	19.0	20.0	19.5	20.0
8	17.5	18.0	19.0	19.0
9	15.5	N/A	17.0	16.0

Table 12: Mean TX output power

4.6 Digital IO Voltage Specifications

Parameters	Description	VDDIO	Min	Max	Unit
V_{IL_VDDIO}	Low input threshold for all GPIO and SDIO pins	3.3	-	1.2	V
V_{IH_VDDIO}	High input threshold for all GPIO and SDIO pins	3.3	1.7	-	V
V_{OL_VDDIO}	Low output voltage for all GPIO and SDIO pins, assuming an 8mA load	3.3	-	0.25	V
V_{OH_VDDIO}	High output voltage for all GPIO and SDIO pins, assuming an 8mA load	3.3	2.9	-	V

Table 13: Digital IO voltage specifications

5 Module Dimensions

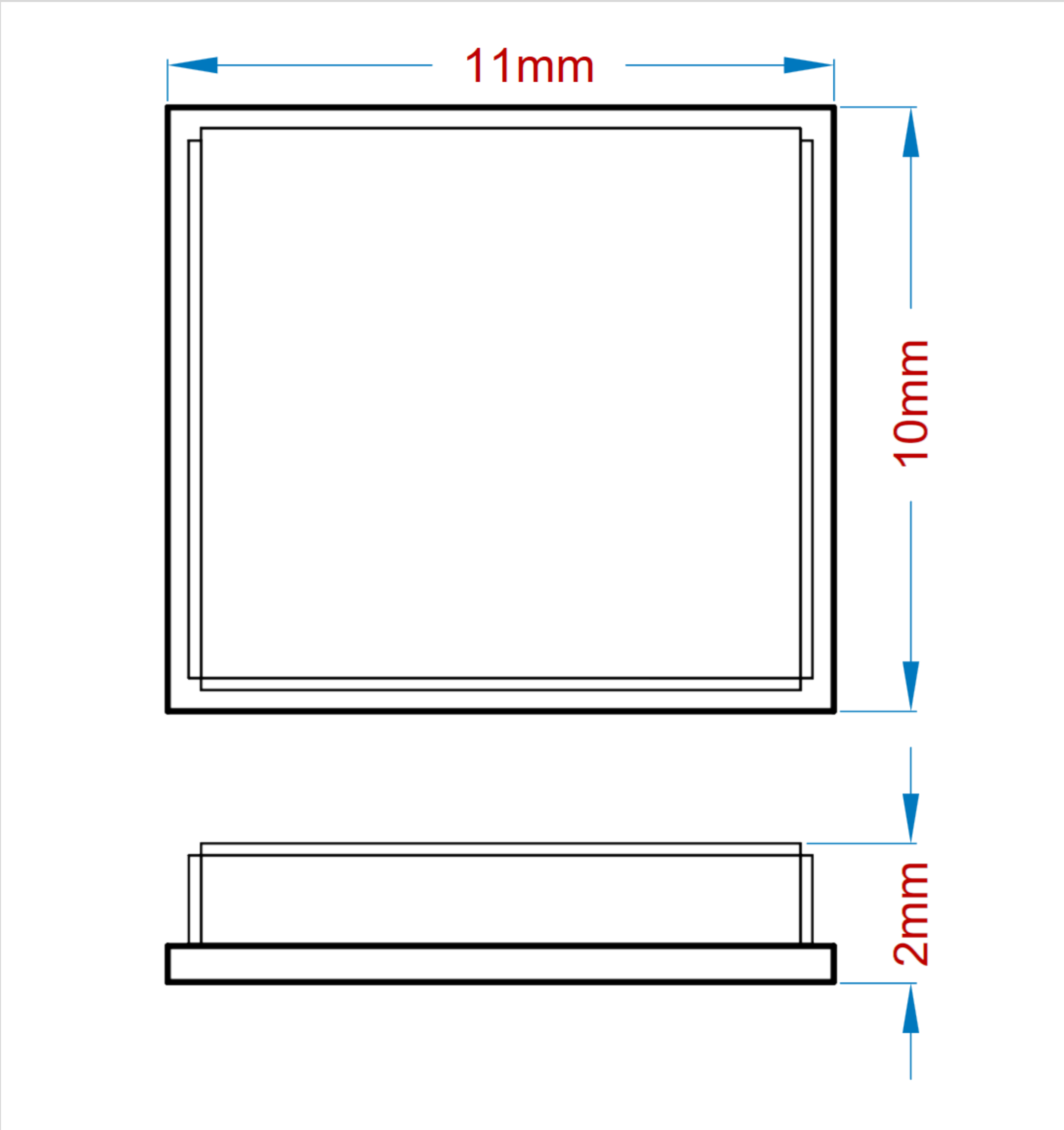

Figure 11: Module dimensions

6 Recommended PCB Footprint

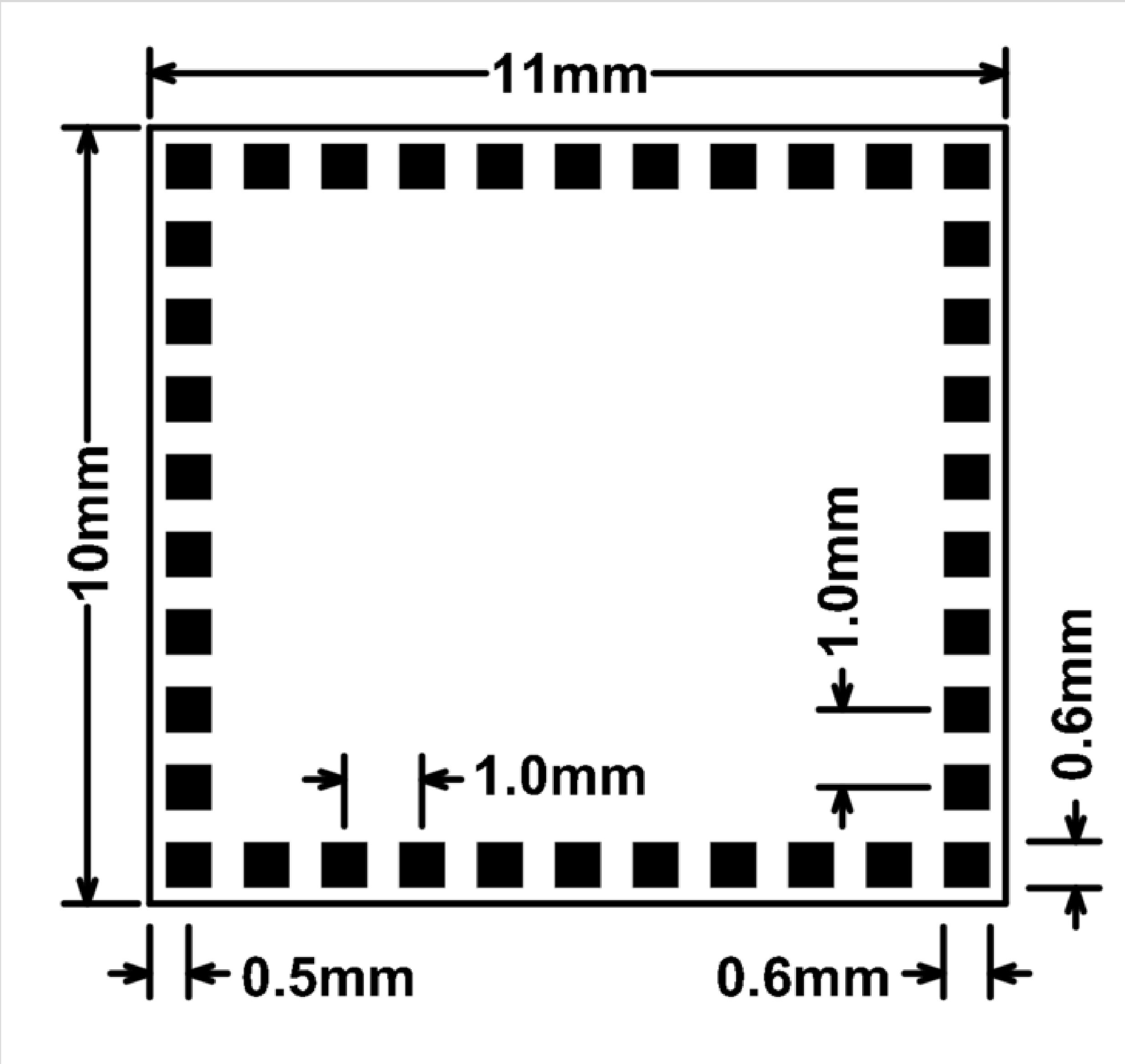

Figure 12: PCB footprint

7 Certification

7.1 FCC

The MM8108-MF15457 has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one of the following measures:

- Reorient or relocate the receiving antenna
- Increase the separation between the equipment and receiver
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected
- Consult the dealer or an experienced radio/TV technician for help

FCC caution: Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate this equipment.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

7.1.1 FCC Radiation Exposure Statement:

This equipment complies with FCC radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with a minimum distance 20 cm between the radiator and your body.

7.1.2 Important Note To Integrators

This module is intended for OEM integrators. It is only FCC authorized for the specific rule parts listed on the grant, and the host product manufacturer is responsible for compliance with any other FCC rules that apply to the host not covered by the modular transmitter grant of certification. The final host product still requires Part 15 Subpart B compliance testing with the modular transmitter installed.

Additional testing and certification may be necessary when multiple modules are used.

7.1.3 End Product User Manual Requirement

In the user manual of the end product, the end user has to be informed to keep at least 20 cm of separation with the antenna while this end product is installed and operated. The end user has to be informed that the FCC radio-frequency exposure guidelines for an uncontrolled environment can be satisfied.

The end user has to also be informed that any changes or modifications not expressly approved by the manufacturer could void the user's authority to operate this equipment.

This device complies with Part 15 of FCC rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference and (2) this device must accept any interference received, including interference that may cause undesired operation.

7.1.4 End Product Label Requirement

The end product must be labeled in a visible area with the following:

Contains FCC ID: 2A74O-737B5B This device complies with Part 15 of FCC rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference and (2) this device must accept any interference received, including interference that may cause undesired operation.

Figure 13: End product label requirement

7.2. IC

This device contains licence-exempt transmitter(s)/receiver(s) that comply with Innovation, Science and Economic Development Canada’s licence-exempt RSS(s). Operation is subject to the following two conditions:

(1) This device may not cause interference.

(2) This device must accept any interference, including interference that may cause undesired operation of the device.

Cet appareil contient des émetteurs / récepteurs exempts de licence qui sont conformes au (x)

RSS (s) exemptés de licence d’Innovation, Sciences et Développement économique Canada. L’opération est soumise aux deux conditions suivantes:

(1) Cet appareil ne doit pas provoquer d'interférences.

(2) Cet appareil doit accepter toute interférence, y compris les interférences susceptibles de provoquer un fonctionnement indésirable de l'appareil.

This radio transmitter 29791-737B5B has been approved by Innovation, Science and Economic Development Canada to operate with the antenna types listed below, with the maximum permissible gain indicated. Antenna types not included in this list that have a gain greater than the maximum gain indicated for any type listed are strictly prohibited for use with this device.

Le présent émetteur radio 29791-737B5B a été approuvé par Innovation, Sciences et Développement économique Canada pour fonctionner avec les types d'antenne énumérés ci-dessous et ayant un gain admissible maximal d'antenne. Les types d'antennes non inclus dans cette liste qui ont un gain supérieur au gain maximal indiqué pour tout type listé sont strictement interdits pour une utilisation avec cet appareil.

7.2.1 IC Radiation Exposure Statement:

This equipment complies with IC RSS-102 radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with a minimum distance of 20cm between the radiator & your body.

Cet équipement est conforme aux limites d'exposition aux rayonnements IC établies pour un environnement non contrôlé. Cet équipement doit être installé et utilisé avec un minimum de 20 cm de distance entre la source de rayonnement et votre corps.

7.2.2 Important Note To Integrators

This module is intended for OEM integrators. The OEM integrator is responsible for complying with all the rules that apply to the product into which this certified RF module is integrated. Additional testing and certification may be necessary when multiple modules are used. Any changes or modifications not expressly approved by the manufacturer could void the user's authority to operate this equipment.

7.2.3 End Product User Manual Requirement

In the users manual of the end product, the end user has to be informed to keep at least 20 cm separation with the antenna while this end product is installed and operated. The end user has to be informed that the IC radio-frequency exposure guidelines for an uncontrolled environment can be satisfied.

The end user has to also be informed that any changes or modifications not expressly approved by the manufacturer could void the user's authority to operate this equipment. Operation is subject to the following two conditions:

(1) this device may not cause harmful interference

(2) this device must accept any interference received, including interference that may cause undesired operation.

7.2.4 End Product Label Requirement

The end product must be labeled in a visible area with the following:

Contains IC: 29791-737B5B

Figure 14: End product label requirement

The Host Model Number (HMN) must be indicated at any location on the exterior of the end product, product packaging, or product literature, which shall be available with the end product or online.

8 Part Numbers and Ordering Information

Part Number	Packing Type	MOQ and Order Multiples	Part Size (mm)	Description
MM8108-MF15457	Tray	100	10 x 11 x 2	IEEE 802.11ah Sub-1 GHz 1/2/4/8 MHz Wi-Fi HaLow Module

Table 14: Part number and ordering information

9 Handling and Storage

The MM8108-MF15457 module is a moisture-sensitive device rated at Moisture Sensitive Level 3 (MSL3) per IPC/JEDEC J-STD-20.

After opening the moisture-sealed storage bag, modules that will be subjected to reflow solder or other high-temperature processes must be:

1. Mounted to a circuit board within 168 hours at factory conditions (≤30°C and <60% RH),

OR

2. Continuously stored per IPC/JEDEC J-STD-033

Modules exposed to moisture and environmental conditions exceeding packaging and storage conditions MUST be baked before mounting, according to IPC/JEDEC J-STD-033.

Failure to meet packaging and storage conditions will result in irreparable damage to modules during the solder reflow process.

10 Revision History

Release Number	Release Date	Release Notes
Version 4	30 Sep 2025	Updated pin descriptions Updated host interface schematic designs
Version 3	18 Sep 2025	Updated formatting Added USB dongle reference design Updated power management Updated timing graphs Updated electrical specifications Updated RF specifications Added digital IO specifications
DS101	28 Feb 2025	Added power on and reset timing details Updated standby power consumption
DS100	8 Jan 2025	Initial release

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